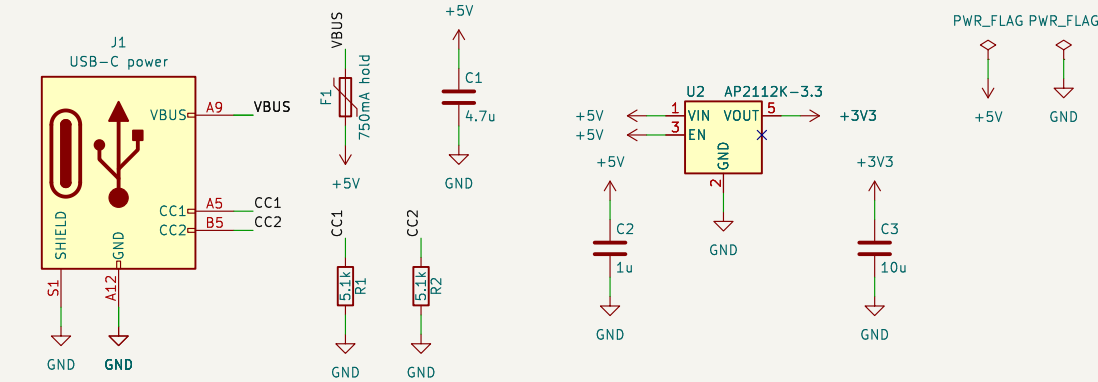
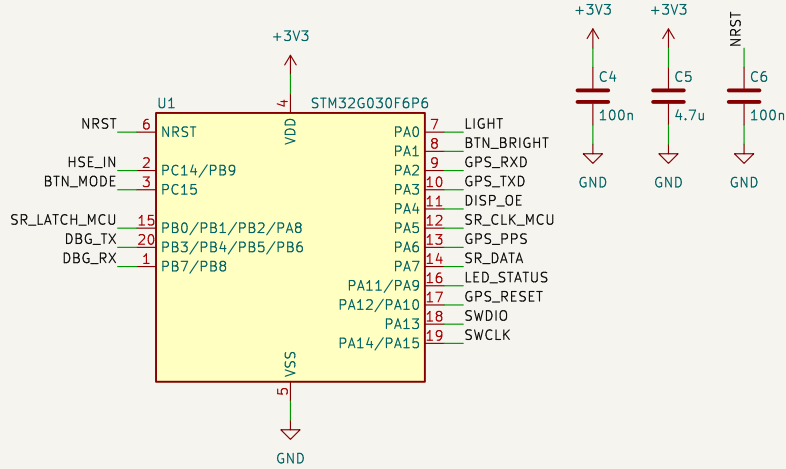


POWER: USB-C 5 V in, power only



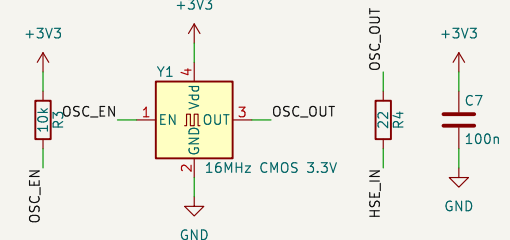
MCU: STM32G030F6P6 (TSSOP-20)

HSE in BYPASS mode on PC14: this package has no OSC_OUT, so no crystal.



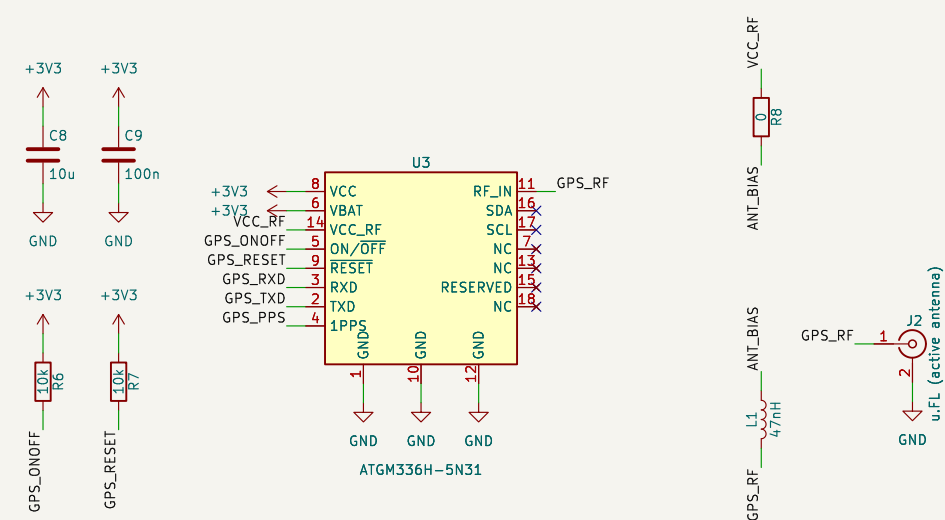
Clock: 16 MHz 3.3 V CMOS oscillator.

Same footprint fits a CMOS-output TCXO (check its pin 1 is EN or NC, not VC).

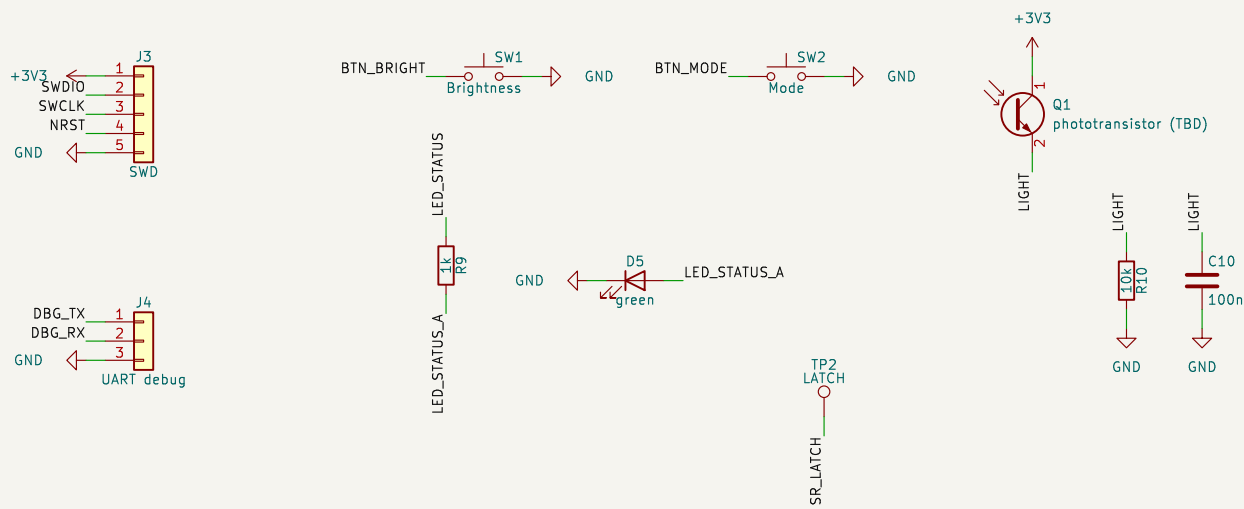


GPS: ATGM336H-5N31

Uses the u-blox MAX footprint (same 18-pad layout). Active antenna via u.FL.



I/O: SWD, debug UART, mode + brightness buttons, status LED, light sensor (optional)



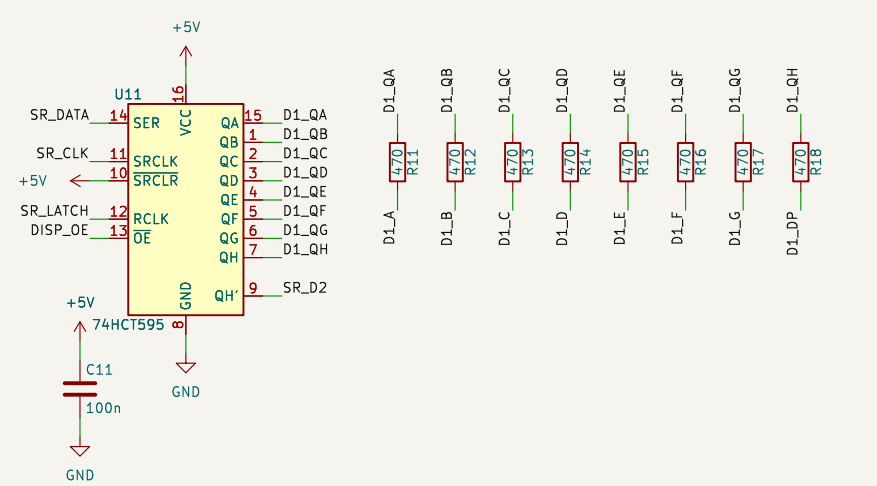
DISPLAY: HH:MM:SS.mmmm, 10 digits on five Kingbright DC56-11 duals, each digit statically driven by a 74HCT595 (5 V).

Chain: SR_DATA -> U11 -> U12 ... -> U20. All latch together on SR_LATCH. /OE = PWM brightness.

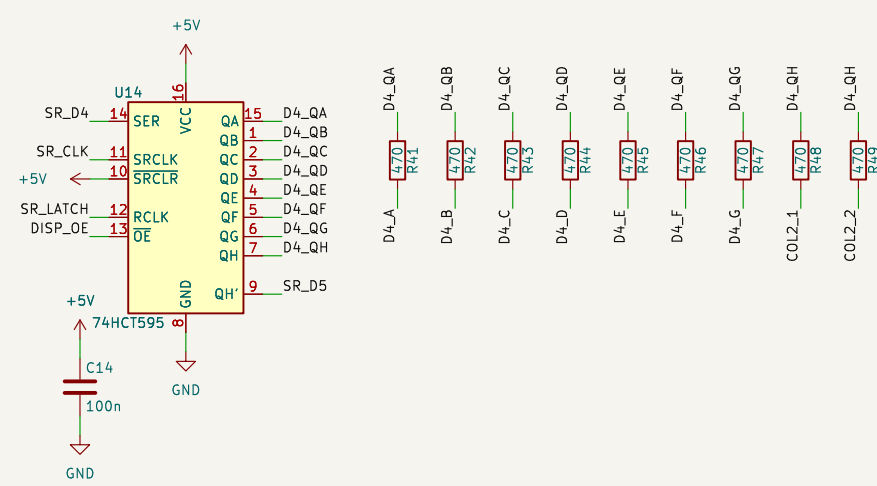
Segment resistors 470R: ~5.7-6.4 mA per LED. 8 LEDs per 595 (~46-51 mA); U12/U14 drive 9 (~51-58 mA). Limit 70 mA.

All 82 LEDs on: ~0.47-0.52 A from 5 V. Firmware must cap /OE duty for an all-segments self-test.

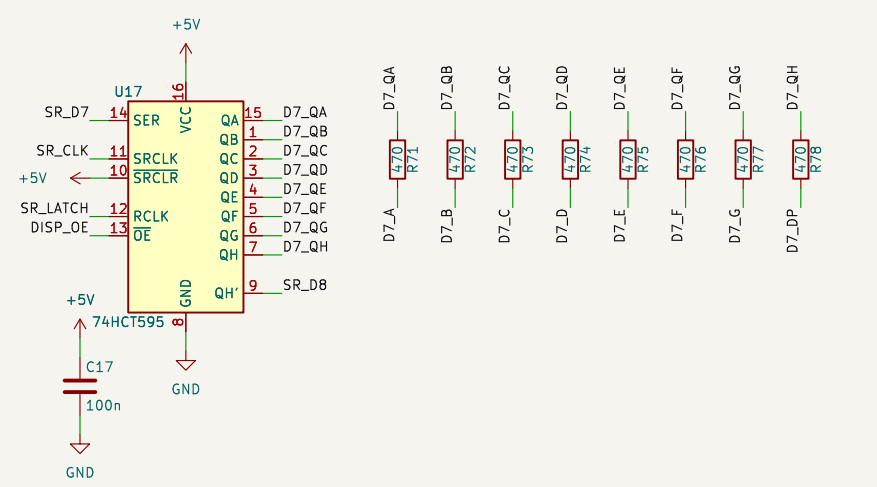
Digit 1: H tens (DS1 DIG1)



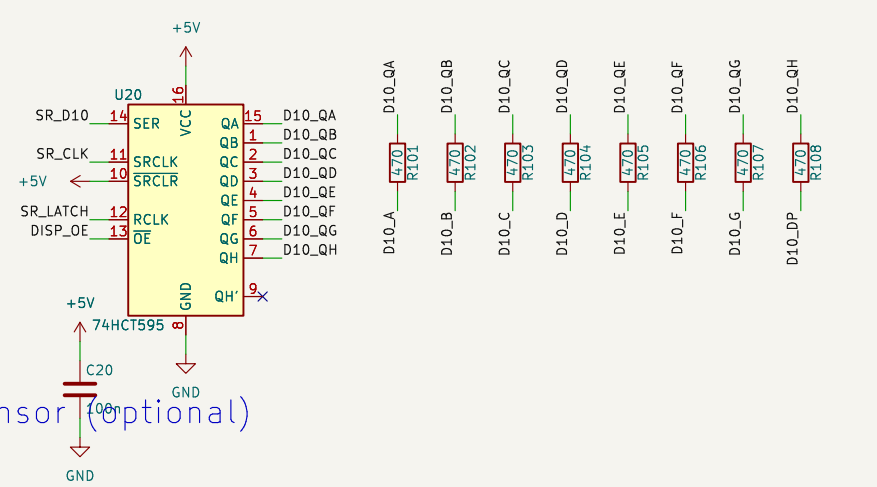
Digit 4: M units (DS2 DIG2)



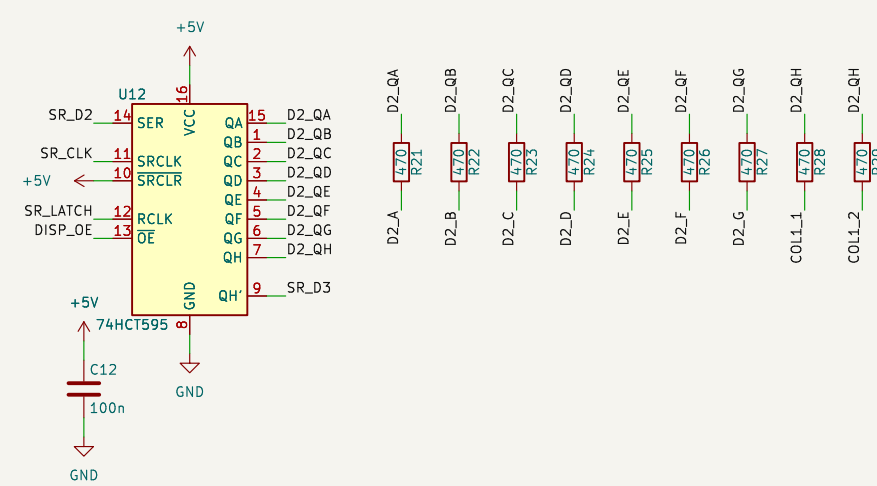
Digit 7: 1/10 s (DS4 DIG1)



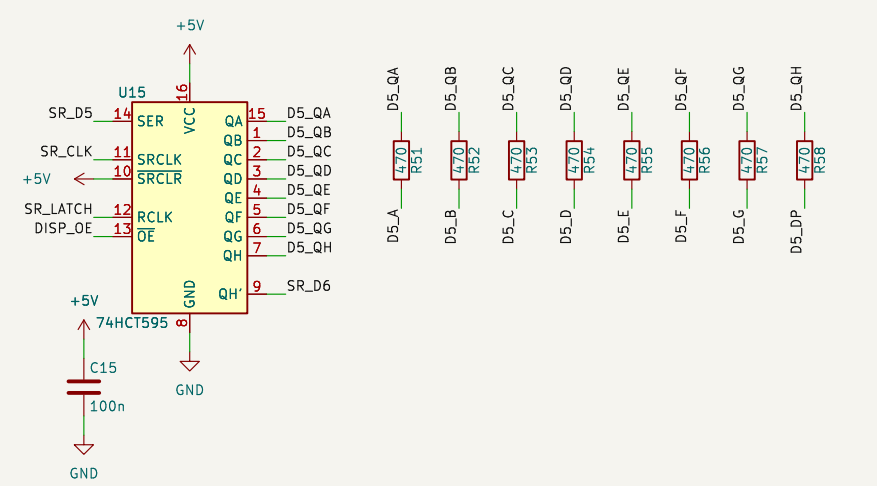
Digit 10: 1/10000 s (DS5 DIG2)



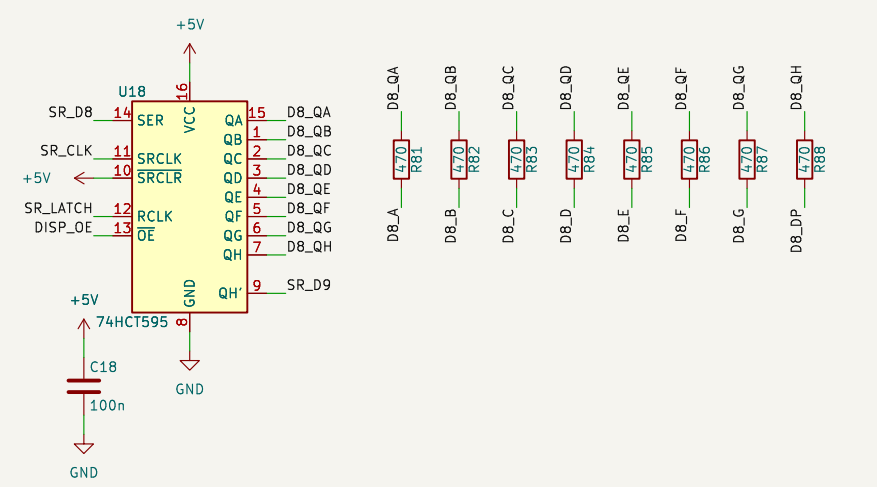
Digit 2: H units (DS1 DIG2)



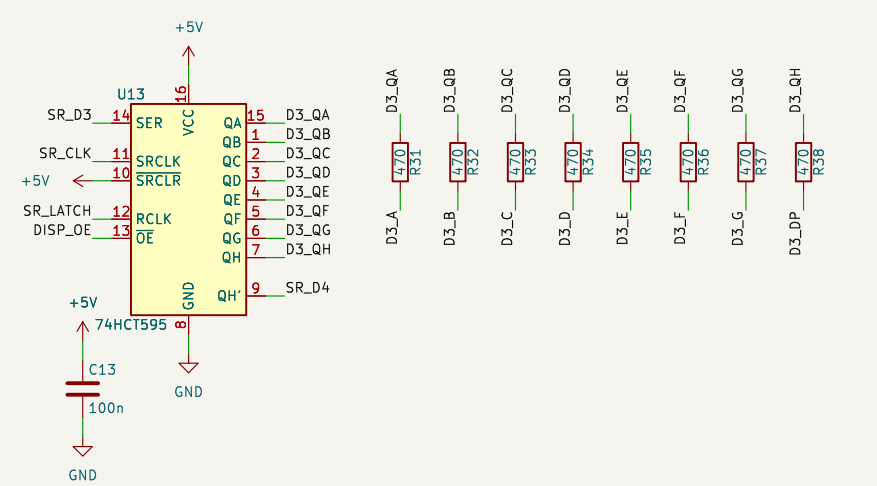
Digit 5: S tens (DS3 DIG1)



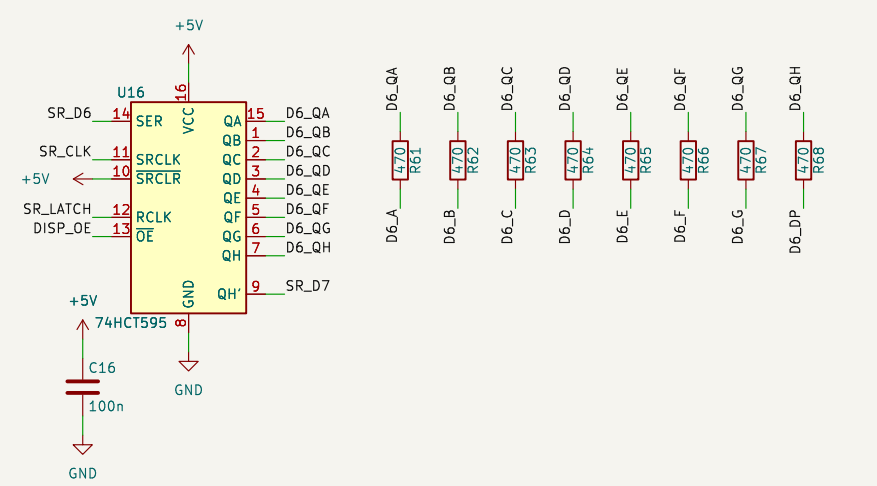
Digit 8: 1/100 s (DS4 DIG2)



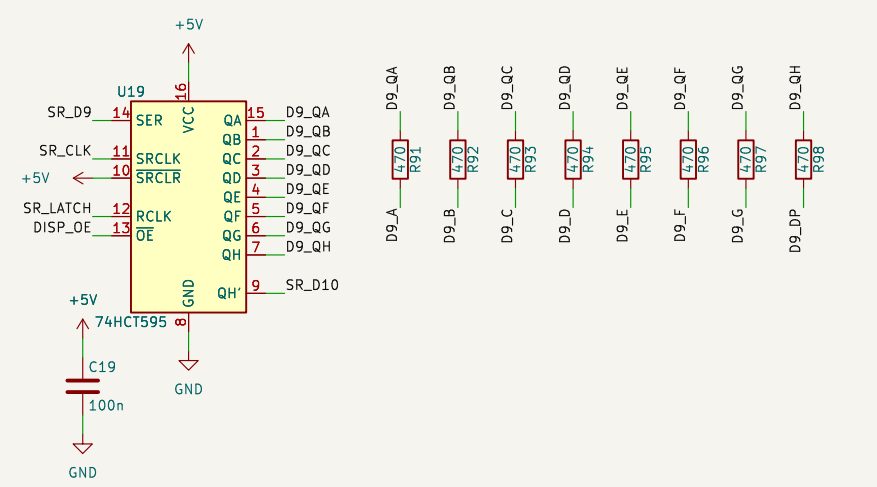
Digit 3: M tens (DS2 DIG1)



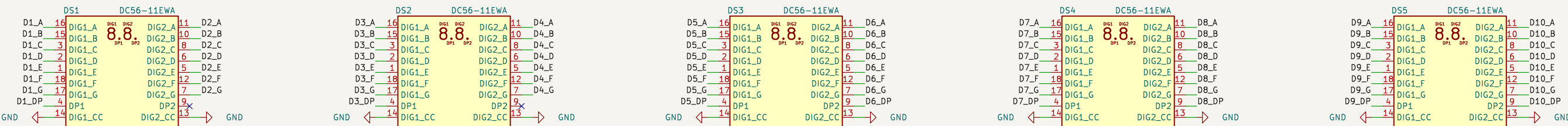
Digit 6: S units, DP = decimal point (DS3 DIG2)



Digit 9: 1/1000 s (DS5 DIG1)



Displays: Kingbright DC56-11EWA, 0.56in dual digit, common cathode, 18 pins (every segment on its own pin).



Colons: between DS1|DS2 (COL1) and DS2|DS3 (COL2). Two 3 mm red LEDs each, driven from QH of U12 / U14.



Generated by hardware/tools/gen_schematic.py – edit the script, not this file

Sheet: /
File: gpsclock.kicad_sch

Title: GPS ms wall clock

Size: A2 Date: 2026-10-06

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 0.2

Id: 1/1